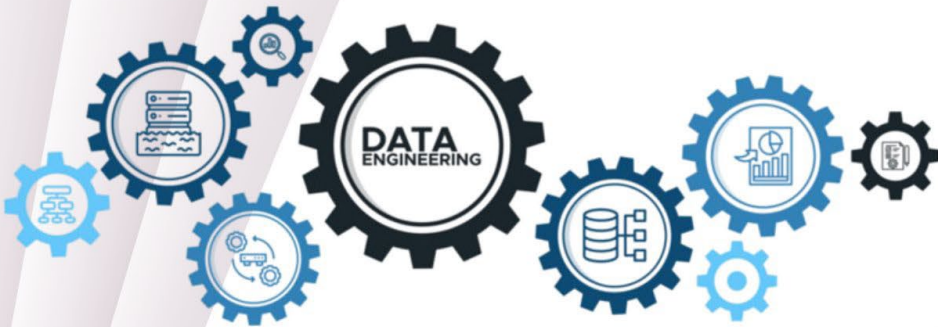


# ESSENTIALS OF DATA ENGINEERING

*Designing and Constructing Reliable  
Data System*



*Dr. Mukesh Saini*

**Essentials of Data Engineering**

by Dr. Mukesh Saini

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## **DEDICATION**

To my parents, family and friends, whose unwavering support and belief in my dreams have been my guiding light. Your love and encouragement have made this journey possible. Thank you for always believing in me.



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# Preface

The genesis of this book traces back to transition from data science to data engineering. As data science gained prominence, companies heavily invested in data science talent with high hopes of significant returns. However, many data scientists frequently encountered fundamental issues that their training did not adequately prepare them for—such as data collection, cleansing, access, transformation, and infrastructure. These challenges are precisely what data engineering seeks to address.

## How This Book Differs

Before diving into what you can expect from this book and how it will benefit you, it's important to clarify what this book does not aim to be. This book is not centered around using specific tools, technologies, or platforms for data engineering. While there are many valuable resources that take this approach, those tend to become outdated relatively quickly. Instead, our focus is on imparting the fundamental concepts that underpin data engineering.

## The Essential Focus of This Book

This book seeks to address a notable gap in existing data engineering literature. While numerous technical resources delve into specific tools and technologies, there remains a challenge in understanding how to integrate these components effectively for real-world applications. The goal is to connect the dots across the entire data lifecycle, illustrating how to seamlessly weave together diverse technologies. This approach enables you to meet the requirements of downstream data consumers such as analysts, data scientists, and machine learning engineers.

The core concept of this book revolves around the data engineering lifecycle: data generation, storage, ingestion, transformation, and serving. Despite the ever-changing landscape of specific technologies and vendor products, these stages have remained fundamentally consistent throughout the evolution of data practices. By embracing this framework, readers will gain a solid foundation for applying technologies to solve real-world business challenges.

The objective is to establish principles that span across two key dimensions. First, to distill data engineering into principles adaptable to any relevant technology. Second, to present principles that endure over time, drawing from lessons learned amidst the technological upheavals of the past two decades. This mental framework remaining valuable for a decade or more into the future.

## Who Will Benefit Most from This Book

Primary audience for this book comprises technical practitioners: mid- to senior-level software engineers, data scientists, and analysts eager to transition into data engineering. It also includes data engineers deeply involved in specific technologies seeking a broader perspective. The secondary audience consists of data stakeholders with technical backgrounds, such as data team leads overseeing teams of data engineers or directors of data warehousing looking to migrate to cloud-based solutions.

Ideally, you're someone driven by curiosity and a desire to learn the very reason you're exploring this book. You actively keep pace with developments in data technologies, regularly reading books and articles on topics like data warehousing, data lakes, batch and streaming systems, orchestration, modeling, and cloud technologies. This book aims to help you synthesize this knowledge, providing a cohesive understanding of data engineering across various technologies and paradigms.

## Prerequisites

Assume readers possess a solid understanding of typical data systems used in corporate settings. Additionally, expect familiarity with SQL and Python, or another programming language, along with some experience using cloud services.

There are abundant resources available for those looking to practice Python and SQL. Numerous free online platforms such as blogs, tutorials, and YouTube offer learning materials, and new Python books are published regularly.

The cloud offers unparalleled opportunities for hands-on experience with data tools and recommend that aspiring data engineers create accounts with cloud providers like AWS, Azure, Google Cloud Platform, Snowflake, or Databricks. Many of these platforms offer free-tier options, although users should monitor costs carefully and start with small datasets and single-node clusters during their studies.

For individuals aiming to familiarize themselves with corporate data systems outside of a formal work environment, challenges exist. This can pose barriers for aspiring data engineers seeking their first job. This book aims to bridge this gap, encourage newcomers to read for broad concepts initially, and refer to the additional resources. On a second pass through the material, focus on unfamiliar terms and technologies. Supplement your

understanding using resources like Google, Wikipedia, blogs, YouTube, and vendor websites to fill any knowledge gaps.

## Enhancing Your Skills: What You'll Learn and How It Will Boost Your Abilities

This book aims to establish a strong foundation for tackling practical data engineering challenges. By the end of this book, you will achieve the following:

- Gain insights into how data engineering influences your current role (whether you're a data scientist, software engineer, or data team lead).
- Navigate through marketing noise to select the most suitable technologies, data architectures, and processes.
- Utilize the data engineering lifecycle effectively to design and construct a robust architecture.
- Implement best practices at every phase of the data lifecycle.

Moreover, you will be equipped to:

- Apply data engineering principles within your current role, whether as a data scientist, analyst, software engineer, or data team lead.
- Integrate various cloud technologies to cater to the needs of downstream data consumers.
- Evaluate data engineering challenges using a comprehensive framework of best practices.
- Integrate data governance and security throughout the data engineering lifecycle.

## Navigating This Book

This book is composed of four parts:

- **Part I**, “Structural Foundation and Essential Building Blocks”
- **Part II**, “Navigating the Data Engineering Lifecycle: A Deep Drive”
- **Part III**, “Safeguarding Data in the Digital Era: Security, Privacy, and the Future of Data Engineering”
- **Appendices A**: covering serialization and compression

In **Part I**, begin by defining data engineering in **Chapter 1**, then map out the data engineering lifecycle in **Chapter 2**. In **Chapter 3**, talk about *good and effective data architecture*.

**Part II** builds on **Chapter 2** to cover the data engineering lifecycle in depth; each lifecycle stage—*data generation, storage, ingestion, transformation and serving* is covered in its own

chapter. **Part II** is arguably the heart of the book, and the other chapters exist to support the core ideas covered here.

**Part III** covers additional topics. In **Chapter 9**, talk about *security and privacy*. While security has always been an important aspect of the data engineering profession, its significance has heightened due to the increase in for-profit hacking and state-sponsored cyber-attacks. As for privacy, the era of corporate indifference is over no company desires to be featured in headlines due to lax privacy practices. Mishandling personal data can lead to severe legal consequences, especially with the implementation of GDPR, CCPA, and other regulations. In summary, prioritizing security and privacy is essential in all data engineering endeavors.

During a journey in data engineering, conducting research for this book and engaging in interviews with numerous experts, extensively contemplated the direction of the field in both the short and long term. **Chapter 10** delves into few speculative thoughts on the future of data engineering. Predicting the future is inherently uncertain, and only time will reveal the accuracy of ideas.

In the appendices, cover a handful of technical topics that are extremely relevant to the day-to-day practice of data engineering but didn't fit into the main body of the text. Specifically, engineers need to understand serialization and compression (see **Appendix A**) as well as additional resources and references (see **Appendix B**).

For instance, in analyzing this book, you might notice certain words appearing frequently. By replacing these common words with shorter tokens during compression, you could potentially achieve a compression ratio of 2:1 or greater. However, sophisticated compression algorithms use advanced mathematical techniques to achieve even higher compression ratios, often reaching 10:1 or more for text data.

It's important to note that these discussions focus on lossless compression algorithms. When data is compressed using a lossless algorithm, decompression results in an exact copy of the original data, bit-for-bit. In contrast, lossy compression algorithms, used in audio, images, and video, aim to preserve perceptual fidelity rather than exact data replication. Data engineers encounter lossy compression in media processing pipelines but typically use lossless compression for analytical serialization, where data fidelity is crucial.

Traditional compression engines like gzip and bzip2 excel at compressing text data and are commonly applied to formats such as JSON, JSONL, XML, CSV, and others. In recent years, a new generation of compression algorithms has emerged that prioritize speed and CPU efficiency over achieving the highest possible compression ratio. Examples include Snappy, Zstandard, LZFSE, and LZ4. These algorithms are widely employed in compressing data within data lakes and columnar databases to optimize for rapid query performance.

---

# Additional Resources and References

This appendix discusses additional resources and references that could be valuable for understanding various aspects of data engineering, data architecture, and related topics.

## Additional Resources

### Books

- *Designing Data-Intensive Applications* by Martin Kleppmann
- *Streaming Systems: The What, Where, When, and How of Large-Scale Data Processing* by Tyler Akidau et al. (O'Reilly)
- *Data Science for Business: What You Need to Know about Data Mining and Data-Analytic Thinking* by Foster Provost and Tom Fawcett (O'Reilly)

### Articles and Papers

- *"Lambda Architecture: Design Simplicity and Performance"* by Nathan Marz
- *"Kappa Architecture: Simplifying Big Data Pipelines"* by Jay Kreps
- *"Event-Driven Architecture: Software that Keeps on Giving"* by Martin Fowler
- *"Data Mesh: A paradigm shift in big data architecture"* by Zhamak Dehghani

### Websites and Blogs:

- *Towards Data Science* ([towardsdatascience.com](https://towardsdatascience.com)) - A platform with numerous articles on data engineering and related topics.
- *The Data Engineering Cookbook* ([dataengineeringcookbook.com](https://dataengineeringcookbook.com)) - Offers practical guides and recipes for data engineering tasks.
- *The Data Warehousing Institute* ([tdwi.org](https://tdwi.org)) - Provides resources, research, and education on data warehousing and BI.

## Documentation and Technical Guides:

- *Apache Kafka Documentation* ([kafka.apache.org/documentation/](https://kafka.apache.org/documentation/)) - Comprehensive documentation for Kafka, a popular distributed streaming platform.
- *Apache Spark Documentation* ([spark.apache.org/documentation/](https://spark.apache.org/documentation/)) - Official documentation for Apache Spark, a unified analytics engine for big data processing.

## Academic Resources

- *IEEE Transactions on Knowledge and Data Engineering* ([tkde.ieee.org](https://tkde.ieee.org/)) - Publishes research articles on data engineering topics.
- *Journal of Data Science and Analytics* ([springer.com/journal/41060](https://springer.com/journal/41060)) - Covers theoretical and practical aspects of data science and analytics.
- These resources cover a wide range of topics from foundational principles to advanced techniques in data engineering and architecture. They are essential for anyone looking to deepen their understanding or stay updated in this rapidly evolving field.

These resources cover a wide range of topics from foundational principles to advanced techniques in data engineering and architecture. They are essential for anyone looking to deepen their understanding or stay updated in this rapidly evolving field

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- *“Modernizing Business Data Indexing”* by Benjamin Douglas and Mohammad Mohtasham
- *“The What, Why, and When of Single-Table Design with DynamoDB”* by Alex DeBrie
- *“The AI Hierarchy of Needs”* by Monica Rogati
- *“Building Analytics Teams”* by John K. Thompson
- *“Data as a Product vs. Data as a Service”* by Justin Gage
- *“Data Engineering: A Quick and Simple Definition”* by James Furbush
- *“Data Teams”* by Jesse Anderson
- *“The Downfall of the Data Engineer”* by Maxime Beauchemin

- *"The Future of Data Engineering Is the Convergence of Disciplines"* by Liam Hausmann
- *"How Creating a Data-Driven Culture Can Drive Success"* by Frederik Bussler
- *"On Complexity in Big Data"* by Jesse Anderson
- *"The Rise of the Data Engineer"* by Maxime Beauchemin
- *"A Short History of Big Data"* by Mark van Rijmenam
- *"Skills of the Data Architect"* by Bob Lambert
- *"The Three Levels of Data Analysis: A Framework for Assessing Data Organization Maturity"* by Emilie Schario
- *"What Is a Data Architect? IT's Data Framework Visionary"* by Thor Olavsrud
- *"Big Data Architectures"* Azure documentation
- *"Bounded Context"* by Martin Fowler
- *"A Brief Introduction to Two Data Processing Architectures—Lambda and Kappa for Big Data"* by Iman Samizadeh
- *"The Modern Data Stack: Past, Present, and Future"* by Tristan Handy
- *"Moving Beyond Batch vs. Streaming"* by David Yaffe
- *"A Personal Implementation of Modern Data Architecture: Getting Strava Data into Google Cloud Platform"* by Matthew Reeve
- *"Principled Data Engineering, Part I: Architectural Overview"* by Hussein Danish
- *"Questioning the Lambda Architecture"* by Jay Kreps
- *"Reliable Microservices Data Exchange with the Outbox Pattern"* by Gunnar Morling
- *"The Rise of the Metadata Lake"* by Prukalpa
- *"Run Your Data Team Like a Product Team"* by Emilie Schario and Taylor A. Murphy
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- *Choosing Open Wisely* by Benoit Dageville et al. *Who's Involved with Designing a Data Architecture?*
- *"Choosing the Right Architecture for Global Data Distribution"* Google Cloud Architecture web page
- *"A Comparison of Data Processing Frameworks"* by Ludovik Santos
- *"Data Architecture: A Primer for the Data Scientist"* by W. H. Inmon et al. (Academic Press)
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- *"Data as a Product vs. Data as a Service"* by Justin Gage

- *"The Data Dichotomy: Rethinking the Way We Treat Data and Services"* by Ben Stopford
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- *"Data Warehouse Architecture: Overview"* by Roelant Vos
- *"Data Warehouse Architecture"* tutorial at Javatpoint
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- *"The Design and Implementation of Modern Column-Oriented Database Systems"* by Daniel Abadi et al.
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- *"Enterprise Architecture's Role in Building a Data-Driven Organization"* by Ashutosh Gupta
- *"Five Principles for Cloud-Native Architecture: What It Is and How to Master It"* by Tom Grey
- *"Functional Data Engineering: A Modern Paradigm for Batch Data Processing"* by Maxime Beauchemin
- *"Google Cloud Architecture Framework"* Google Cloud Architecture web page
- *"How to Beat the Cap Theorem"* by Nathan Marz
- *"How to Build a Data Architecture to Drive Innovation—Today and Tomorrow"* by Antonio Castro et al.
- *"How TOGAF Defines Enterprise Architecture (EA)"* by Avancier Limited
- *"Introducing Dagster: An Open Source Python Library for Building Data Applications"* by Nick Schrock
- *"The Log: What Every Software Engineer Should Know About Real-Time Data's Unifying Abstraction"* by Jay Kreps
- *"Microsoft Azure IoT Reference Architecture"* documentation
- Microsoft's *"Azure Architecture Center"*
- *"Modern CI Is Too Complex and Misdirected"* by Gregory Szorc

## Symbols

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## About the Authors

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